

Phase-2 Submission – Data Analytics

Student Name: S.S.TEJASWINY

Register Number: 822423104081

Institution: MRK INSTITUTE OF TECHNOLOGY

Department: COMPUTER SCIENCE AND ENGINEERING

Date of Submission: 05-05-2025

GitHub Repository Link: <https://github.com/Tejaswiny-2005/NM-PROJECT.git>

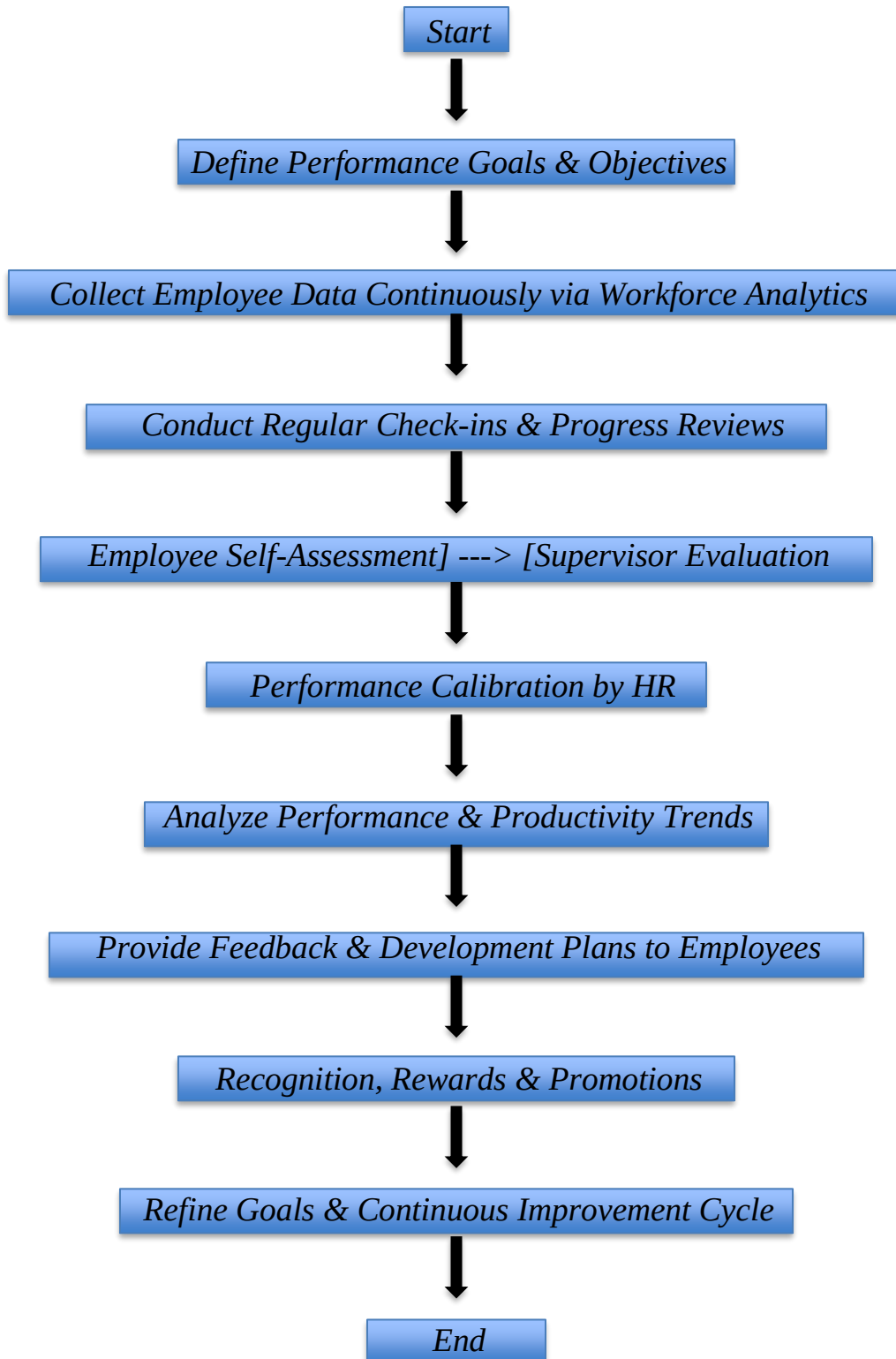
1) Problem Statement

- *Subjectivity and Bias.*
- *Overemphasis on Quantitative Metrics.*
- *Inefficient Feedback Systems.*
- *Lack of Comprehensive Data.*
- *Resistance to Change.*
- *Data Integration Challenges.*

2) Project Objectives

- *Objectively measuring individual and team performance using data from various sources such as attendance, output, engagement, and feedback.*
- *Identifying productivity drivers and bottlenecks to optimize processes and resource allocation.*
- *Detecting skill gaps and training needs to support targeted employee development.*
- *Monitoring workforce trends like turnover, absenteeism, and engagement to improve retention and job satisfaction.*
- *Reducing bias in performance evaluations by basing assessments on consistent, objective data rather than subjective perceptions.*
- *Providing actionable insights for managers and executives to foster a culture of continuous improvement and support strategic workforce planning.*

3) Flowchart of the Project Workflow



4) Data Description

Here is a dataset suitable for evaluating employee performance and productivity trends using workforce analytics:

- **Dataset Name:**
IBM HR Analytics Employee Attrition and Performance
- **Source:**
Kaggle (commonly referenced and widely used in HR analytics research)
- **Data Type:**
Tabular (CSV format)
- **Number of Rows:**
1,470
- **Number of Columns:**
35
- **Static or Dynamic:**
Static dataset (snapshot of employee data at a point in time)
- **Key Fields/Attributes:**
 - *Age*
 - *Gender*
 - *Job Satisfaction*
 - *Environment Satisfaction*
 - *Education Field*
 - *Job Role*
 - *Income*
 - *Overtime*
 - *Percentage Salary Hike*
 - *Tenure (Years at Company)*
 - *Training Time*
 - *Years in Current Role*
 - *Attrition (target variable: Yes/No)*
 - *Performance Rating*
 - *Relationship Status*
 - *Department*

This dataset provides a comprehensive view for analyzing employee performance, productivity, and attrition trends using workforce analytics.

5) Data Preprocessing

Each data processing step-handling missing values, removing duplicates, formatting, encoding, outlier treatment, and documentation-is essential for reliable, unbiased, and interpretable workforce analytics.

1) *Handling Missing Values*

Options:

- *Remove rows or columns with missing values if the missing rate is high and data loss is acceptable.*
- *Impute missing values using mean, median, or mode for numerical data, or use forward/backward fill for time series.*
- *Advanced methods: KNN imputation, regression, or iterative imputation for more accurate filling.*

Reason:

- *To reduce bias, ensure model accuracy, and prevent errors during analysis or machine learning.*

2) *Removing duplicates*

Options:

- *Use functions like drop_duplicates() to eliminate repeated rows.*

Reason:

- *Duplicates can skew analysis and model performance by over-representing certain data points.*

3) *Formatting and parsing data*

Options:

- *Standardize column names (e.g., all lowercase, no spaces).*
- *Convert data types (e.g., strings to dates, integers to floats as needed).*
- *Merge or concatenate datasets carefully, ensuring alignment of columns and handling mismatches.*

Reason:

- *Consistent formatting ensures smooth downstream processing and accurate analysis.*

4) *Encoding categorical variables*

Options:

- *Label encoding for ordinal data.*
- *One-hot encoding for nominal data (e.g., using `pd.get_dummies()`).*

Reason:

- *Machine learning algorithms require numerical input; encoding converts categories into usable numbers.*

5) *Identifying and Treating Outliers*

Options:

- *Visualize with boxplots or use statistical methods (e.g., Z-score, IQR) to detect outliers.*
- *Treat by capping, removing, or transforming outlier values.*

Reason:

- *Outliers can distort statistical summaries and model predictions, so addressing them improves robustness.*

6) *Documenting All Transformations*

Options:

- *Maintain a log or script of each transformation step, including the method and rationale (e.g., “Imputed missing ‘Age’ values with median due to right-skewed distribution”).*

Reason:

- *Ensures reproducibility, transparency, and easier debugging or auditing of the data pipeline.*

6) Exploratory Data Analysis (EDA)

- **Univariate Analysis:**

- *Univariate analysis examines individual variables to understand their distribution, central tendency, and spread. For workforce analytics, this could include:*
- *Headcount: Total number of employees.*
- *Tenure Distribution: How long employees have been with the company.*
- *Performance Ratings: Distribution of employee performance scores.*
- *Absenteeism Rate: Frequency of employee absences.*
- *Employee Engagement Scores: Distribution of engagement survey results.*

These metrics provide a foundational understanding of workforce composition and highlight outliers or trends in single variables.

- **Bivariate/Multivariate Analysis:**

1. Bivariate Examples:

- *Performance vs. Tenure: Analyzing if longer tenure correlates with higher performance.*
- *Engagement vs. Turnover: Investigating if low engagement scores are linked to higher turnover rates.*
- *Salary vs. Performance: Assessing whether higher salaries correspond to better performance.*

2. Multivariate Examples:

- *Predictive Modeling: Using regression or machine learning to predict turnover risk based on factors like tenure, engagement, salary, and performance.*
- *Diversity Impact: Exploring how diversity (age, gender, role) affects productivity and retention.*

- **Analysis of key metrics or KPIs :**

- *Employee Productivity: Revenue per employee, output per hour.*
- *Retention Rate: Percentage of employees staying over a period.*
- *Turnover Rate: Percentage of employees leaving.*
- *Absenteeism Rate: Frequency of unscheduled absences.*
- *Engagement Scores: Results from employee surveys.*
- *Quality of Hire: Performance and retention of new hires.*
- *Training ROI: Impact of training on performance metrics.*
- *These KPIs help organizations track workforce health, efficiency, and alignment with business goals.*

- **Summary of insights and patterns identified:**

- *Univariate analysis reveals basic trends, such as high absenteeism or uneven performance ratings, which may signal underlying issues.*
- *Bivariate/multivariate analysis uncovers relationships-such as low engagement leading to higher turnover, or strong correlation between training participation and improved performance.*
- *KPI tracking enables ongoing monitoring of workforce effectiveness, highlighting areas for intervention (e.g., high turnover in specific departments or productivity dips after organizational changes).*
- *Overall, combining these analyses helps organizations make data-driven decisions, optimize talent management, and proactively address workforce challenges.*

7) Tools and Technologies Used

- **Programming Language:**

- **Python:** Widely used for data analysis, machine learning, and automation.
- **R:** Excellent for statistical analysis and data visualization.
- **SQL:** Essential for querying and manipulating HR databases.

- **Notebook/IDE:**

- **Jupyter Notebook:** Interactive environment for Python and R, ideal for exploratory analysis.
- **RStudio:** Dedicated IDE for R, popular in statistical analytics.
- **VS Code / PyCharm:** Versatile IDEs for Python development.

- **Libraries:**

- **For Python:**
 - *pandas:* Data manipulation and analysis.
 - *numpy:* Numerical computations.
 - *matplotlib & seaborn:* Data visualization.
 - *statsmodels & scipy:* Statistical modeling and tests.
 - *openpyxl:* Excel automation.
 - *pyworkforce:* Workforce scheduling and optimization.
- **For R:**
 - *dplyr, tidyr:* Data wrangling.
 - *ggplot2:* Visualization.
 - *caret:* Machine learning.

- **Optional Automation Tools:** pandas-profiling]
 - *Apache Airflow: Workflow scheduling and automation.*
 - *Zapier / Make: Automate HR data flows between apps.*
 - *Python scripts with scheduled tasks (cron/Task Scheduler) for regular reporting and alerts.*

8) Team Members and Contributions

Name	Contribution
SOWNDHARYA.S	✓ <i>Flowchart of the Project Workflow</i> ✓ <i>Data Description</i>
SOWMIYA.G	✓ <i>Problem Statement</i> ✓ <i>Project Objectives</i>